

Detailed Summary

"*Magic Metrics: The Key to Harry Potter's Success*" by Kevelly Batista Diaz (December 2024), investigates the factors driving the massive cinematic success of the *Harry Potter* film franchise. The franchise spans eight movies (released between 2001 and 2011), two of which achieved over \$1 billion globally at the box office.

Objective & Hypothesis

The primary goal was to explore how metrics like the number of unique characters, production budgets, and global IMDb audience ratings relate to box office revenue. It was initially hypothesized that the number of unique characters would *not* have a significant relationship with the franchise's commercial or critical success.

Data Preparation & Exploratory Data Analysis (EDA)

A merged dataset (hp_merged) was created by combining a Kaggle *Harry Potter* dataset with non-commercial IMDb movie ratings and titles datasets. After removing special characters, standardizing variables, and dropping unnecessary columns (like character houses or runtime), the final dataset consisted of 11 columns and 6,432 rows.

- **Distribution and Outliers:** EDA revealed that both box office revenue and budget data were skewed, with most movies clustering in the lower financial tiers. Audience ratings generally sat between 6.6 and 7.7, though a noticeable dip occurred in the later films.
- **The Outlier:** Using z-scores, "*Harry Potter and the Deathly Hallows Part 2*" was identified as a distinct statistical outlier due to its exceptionally high box office earnings coupled with a relatively low IMDb rating. It was kept in the dataset because of its value to the overall exploration.

Statistical Findings

Using Pearson's correlation and p-value hypothesis testing, the study yielded surprising insights:

- **Revenue vs. Ratings:** There is an incredibly strong *negative* correlation (-0.95) between box office revenue and global IMDb ratings, meaning the highest-grossing films actually scored lower with audiences.
- **Budget vs. Revenue:** A moderate positive correlation (0.59) confirmed that larger budgets generally yield higher financial success.

- **Character Count:** The number of unique characters had a very weak positive correlation (0.21) with box office performance and weak negative correlations with audience ratings (-0.28) and budget (-0.30).
- **Hypothesis Conclusion:** Despite some correlations being weak, the p-values for all tested relationships were exceptionally small (approaching 0.0), proving that these connections are statistically significant and not due to random chance. Consequently, the initial null hypothesis was rejected.

It was concluded that the franchise's cinematic success is complex and likely driven by external factors such as marketing and financial investment rather than narrative traits like character diversity. Future research suggestions include applying linear regression models and comparing *Harry Potter* to other literary-to-film adaptations.